Reverse Proxy and deployemnt of the project

AWS

AWS is Amazon's cloud services

It let's you

1. Rent servers
2. Manange Domains
3. Upload objects (mp4 files, jpgs, mp3s, etc..)
4. Autoscale servers
5. create k8s clusters (kubernetes clusters)

The offering we will be focussing on today is `Renting servers`

It offers various things out of which **EC2 is the major one** which stands for **Elastic compute version 2**

💡 **What is elastic compute ??**

-> Just see the IntrotoDevos notes where we have understood the vm (Hardware -> Hypervisor -> VM concept)

- **Elastic ->** means it can change(increase/decrease the size of the machine) according to the need
- **Compute ->** Operations being performed on the hardware

and Version 2 is simply the version of this service given by AWS

Steps to create your own VM in aws

VMs on AWS are called as `EC2 Server`

Just go to the AWS dashboard and then in the search bar, write **Ec2** and you will see the option to create `EC2` click on it and then

- go to the `launch instances` button in the right side panel

give the name to the instance under the `Name and tags` input (basically name it your project)

- select your OS (operating system)

most of the time Ubuntu is selected

There are also some cases in which you have to change the version of some os accroding to the need. one of them is

- Judge0 -> used by coding platform like leetcode to judge the problem and handle it (it actually has different C groups(**basically you want to ,limit the resource when you are building some thing like leetcode so that code submission does not takes too much time to get judged and these groups help in this**) which run only on Ubuntu version 20) so although Ubuntu has version 24 latest then also you have to shift to a later version to use in your project using Judge0.

- now the next important option to choose from is the **instance type** which simply means which type of machine you want ??

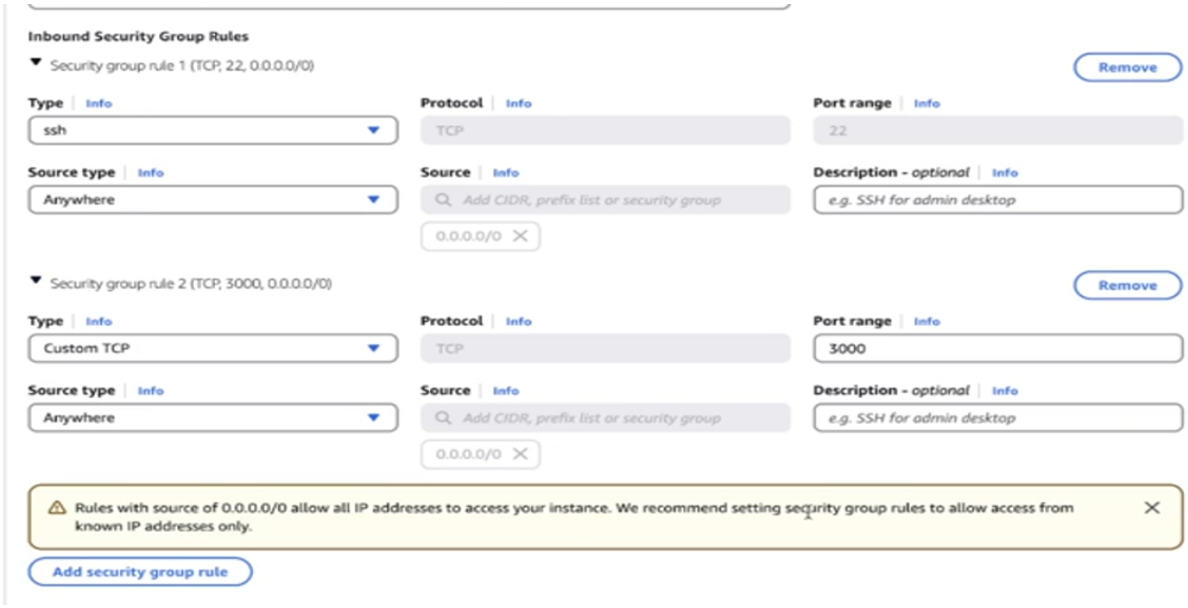
> [!NOTE] **t2-micro is free** but keep in mind that if you have a **react or next.js project** then avoid renting the t2-micro machine, as they take a lot of memory so **you can serve react or next.js application but not build it**
>
> **Build the application and then push, dont build on the machine otherwise you will get memory exceeded so convert the tsx files to html, css and js files and then build and push it**

- Next option is **keypair creation** (same theory as that present in the IntrotoDevops notes) clicking on this will let you give the option of various algorithm through which you can generate keypairs. (basically public key cryptography techniques)[how the keypair will look like] creating this will download the keypair and now you can give this file to anyone whom you want to give access to the machine.

- Now comes the **Network settings** which is important

    - **Security groups** -> By default, when you create a vm machine, you cannot visit it although you have used the public ip of this machine (**as the ports block them and you have to open it in order to visit it**)[Security groups lets you do this]
    - so creating the Security group give its name and desc. according to yourself. Now **you should be opening port 22 atleast for yourself to get into this machine with the help of private key** but if you close it, then you are being too safe (useful in the case when you have given someone else private key)
    - Change -> Source type -> Anywhere (you can even restrict to your ip -> benefit (even if your private key gets leaked, then also this port will be able to open from your ip address only, it is that strict))
    - Now despite the above security groups, you will also have to add other security groups if you want to make some more stricter rules
    - As you are going to run **Http server** which under the hood uses **TCP** so select `custom TCP`(set it to `All traffic` if you want any protocol to come) and then give the port -> `3000` or you can give the port range `3000-3002` (**given in the case when you have multiple projects running in the same machine**)
    - For port `3000` you want that anyone can visit my website so select Source type -> Anywhere (but here **it should be this option**) not like the above ki port `22` pe v koi v aa jaye(ideally upar wale me `My ip` selected hona chahiye tha so that port `22` sirf tm access kr pao although anyone else have even private key and port `3000` se koi v access kr paye tmhare website ko) [you can also

add here `Custom` option if you want ki koi specific ips pe he chle tmhari website then you can add a list of ips here -> Usecase -> you want bas office ke andar chle so put all the possible ip address of your office]



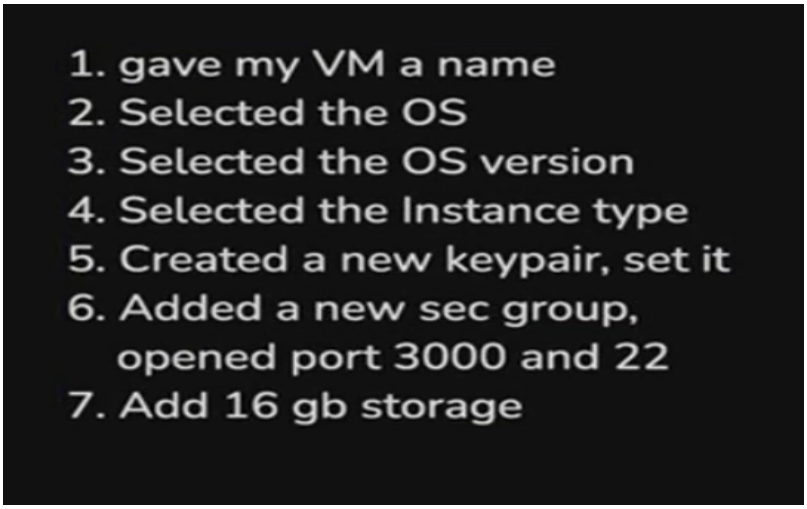
You can also add as many `Security group` as you want by clicking on the `Add Security group rule`. Whatever we have discussed above can be seen in the above pic

💡 Why your vm machine is kept close by default ??

-> You must be wondering why vm machine is kept close from internet by default and you are given so much options regarding the network security then below is the reason :-

- **Security ->** Make your vm machine more secure and you know and can control the security
- **maybe you are running some service providing platform ->** you might be running a video transporting service that might pick a `.mp4` file and convert it to `360p`, `720p` and `1080p` so for this **Transcoding service**, you **dont need to have internet accessible to this machine**. Another example can be is **Model training** (you basically build a heavy gpu inside your house and **no one is going to look at your progress and hence it is not a good way to make it internet accessible hence keep all the port close by default**)

To summarise, below are the steps which we have performed till now in the AWS :-



Steps to connect to your created VM instance

Step 1 -> In the terminal, Give ssh key permissions

```
chmod 700 kirat-class.pem   // .pem is the file which will have your key pair
```

Step 2 -> Run the ssh into machine

```
ssh -i kirat.pem ubuntu@ip_of_your_machine
```

If You will run the above command without doing the Step 1 you will see an Error something like the below

Every file and folder has some permission (you will see it something in the form `rw---ewwe`)

Permission 0644 -> Now the `644` means each bit is in decimal form of binary represntation

So you have to change the persmission for the pem file to `700` -> as this will give you only permission to read, write and execute [This change is to be done for the first time only to set this for all future `.pem` file]. so to remove the above error just do

```
chmod 700 kirat.pem
```

That's what was the meaning of the Step 1

Step 3 -> Clone the repo

```
git clone your_project_link_which_you_want_to_deploy
```

Step 4 -> Now if you will see the folder by doing `ls` you will see your project there

Step 5 -> Now go inside the project by doing `cd project_name` and then as you know to start the project you first have to do

```
npm install
```

But running the above command will give you error

Reason _>

> [!NOTE] `git` does come pre-installed in Ubuntu, thats why you were able to do `git clone` But `npm` does not come by default in Ubuntu and hence gave error

So to install `node.js` in Ubuntu or basically your vm instance run the command

```
curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.39.3/install.sh | bash
```

`https://raw.githubusercontent.com/nvm-sh/nvm/v0.39.3/install.sh`, **This is basically the link consisting of all the files for `node.js` if you go and visit it, you will see it**

`| bash` -> redirecting to the bash shell (or command)

doing the above thing will give you a message to close and reopen or run the below to continue without repoening

```
export NVM_DIR="$HOME/.num"

OR

source ~/.bashrc
```

so just run any of the above command and you are good to go and now you have `node.js` initialized on your vm

Step 6 -> Now specify which version of `node` you want to install by giving

```
nvm list-remote  // will give you list of all the node version you can install

// to install the latest version just do below
nvm install --lts  // lts are you can say final version of latest BETA so its advisable to install just 1 version less of lts version so that you get the most stable as well as latest from the public release point of view
```

and now if you do `npm -v` and you will see the version of the `node.js` in your vm if not then debug

Step 7 -> Now that you have `node` so now install all the dependencies

```
npm install
```

`ls` will make you see `node_modules` folder inside your folder

Step 8 -> Run the project by doing

```
node index.js
```

and that's it you have **deployed your project** -> Go to the ip address

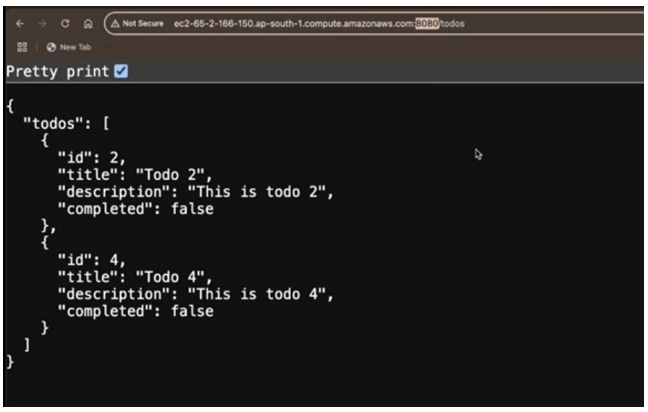
> [!IMPORTANT] Make sure your project code has same or one of the port which is listed in the Network security rules if not, then add the project port in the Network security rules or make changes in the port of the project

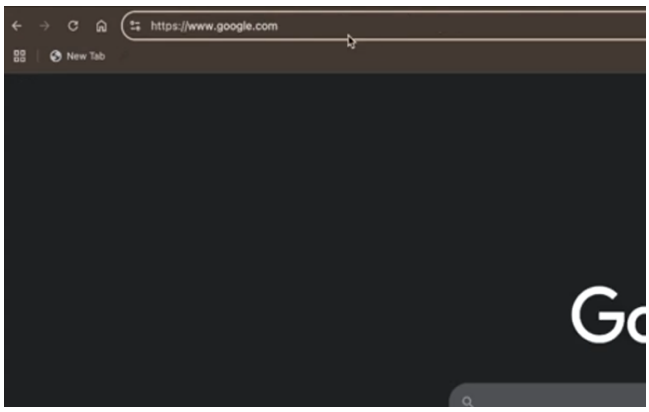
Stiļl there are many things to be done ->

1. Domain name -> ip address look very ugly
2. It is `not-secure`, it should be `https-secure`

Reverse Proxies

if you see yours url vs google one



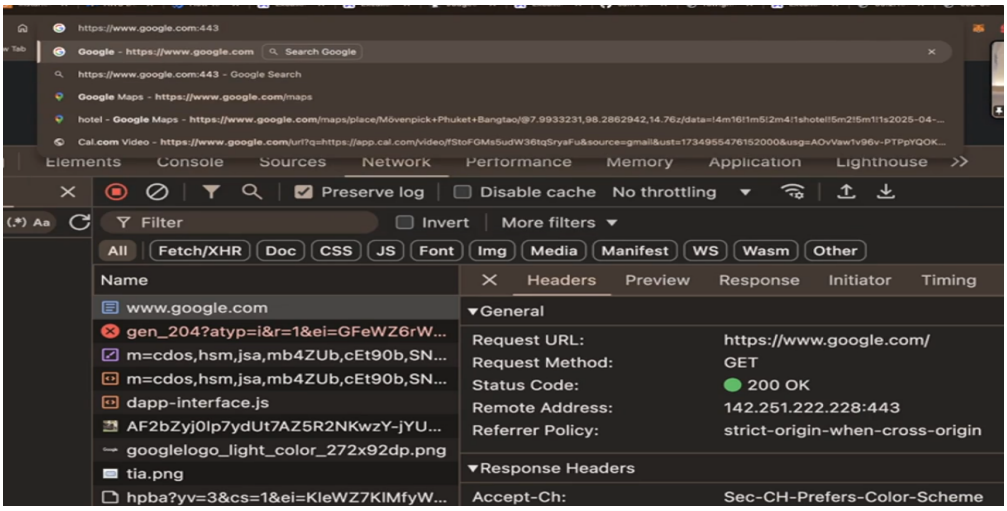


You will notice that `google` does not have **Port no. in its URL** so how they are getting rid of this

Reason for it ->

> [!IMPORTANT] If you are using `https`, then **its default port is 443** and if you are using `http`, then **its default port is 80**

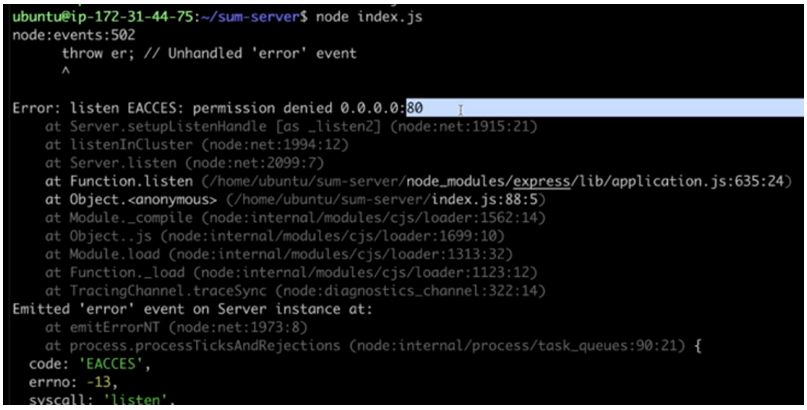
so it is not like that if you go to `google.com`, **port is not present there, It is assumed as you are hittijng the `https` protocol so its port is 443** for `http` protocol if you dont want the port to be shown in the url use **PORT 80**



You can see the Remote address has PORT = 443 under the hood, request are going on this ip and hence **if you are using `https` protocol, you dont have to EXPLICITLY address the port as it has by default 443 port**

The same is true for our website (as it does not have `https` protocol) but as discussed above in Important part, for `http` **PORT 80** works the same as that in `https` **PORT 443** so to make your website get rid of **PORT giving**, **One way can be**

1. **Change the port present in your project to 80 value** and that's it you think this will work BUt you will get some thing like the below



Reason -> OS restricts the use of default port for various protocol, although there are ways to overpass even the OS decisions (learn about `sudo` command)

Even if you were able to run the website at **PORT 80**, then **one very important problem will come**

- **CANT START MULTIPLE APPS ON THE SAME PORT**

> [!CAUTION] **Two processes can't have same port**
>
> will lead to **PORT Conflict**

We need something like -> if port 80 pe request aa rhi from lets say `clothes.com` then it should go to `clothes website` of yours and if port 80 pe request aa rhi h from `shoes.com` then it should go to `shoes website` of yours

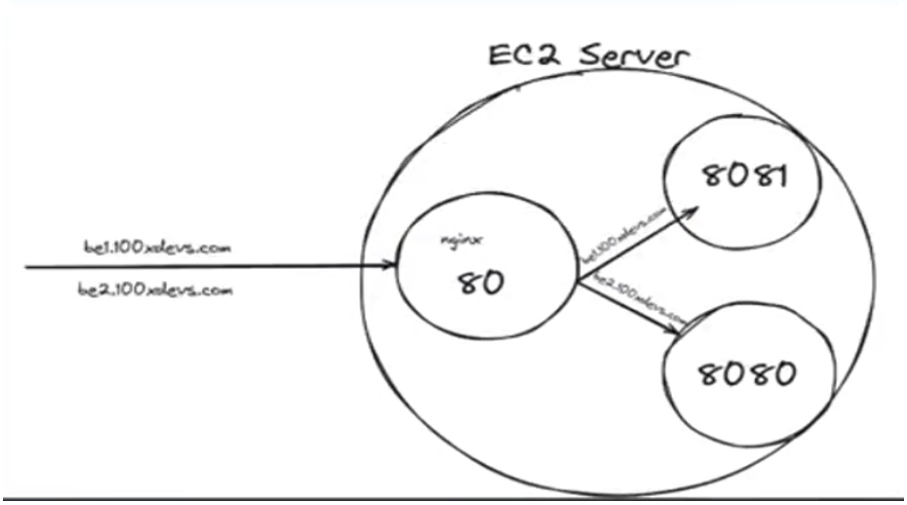
basically we need **REVERSE PROXY**

What is proxy ??

-> You can understand this as a **Mediator to connect between the two network**. **VPN** is one of the example
-> you **forward** your request to some server which has access to that restricted content and through that server you are able to see the restricted content (this is called as **Forward Proxy**)

Now coming to **What is reverse proxy ??**

-> Instead of user going on the seperate `3000` and `3001` port or running your project on the seperate port, you are **introducing nginx in the middle** which will work as the **Middleman**, [`nginx` **runs at port 80** and it decides where to send the request i.e. -> if it coming from `clothes website` then it will forward it to the **port 8080** and if coming from `shoes website` then it will forward it to the **port 8081**]



How this is reverse proxy ??

-> A user is sending the request to `nginx` and that is being **proxied to the different ports according to the reqeust from the website it came from**

Its not actually fully reverse but still it known as reverse proxy.

Some of the other reverse proxy are `traefic`, `HA Proxy`, `Apache`

Nginx

Link to documentation -> Nginx

NGINX is open source software for web serving, **reverse proxying** , caching, load balancing, media streaming, and more. It started out as a web server desgined for maximum performance and stability. In addition to its HTTP server capabilities, NGINX can also function as a proxy server for email (IMAP, POP3, and SMTP) and a reverse proxy and load balancer for HTTP, TCP and UDP servers.

> [!NOTE] Dont think that extra hop aane se **slow ho jayega NGINX, it will but not to that extent which you are thinking**

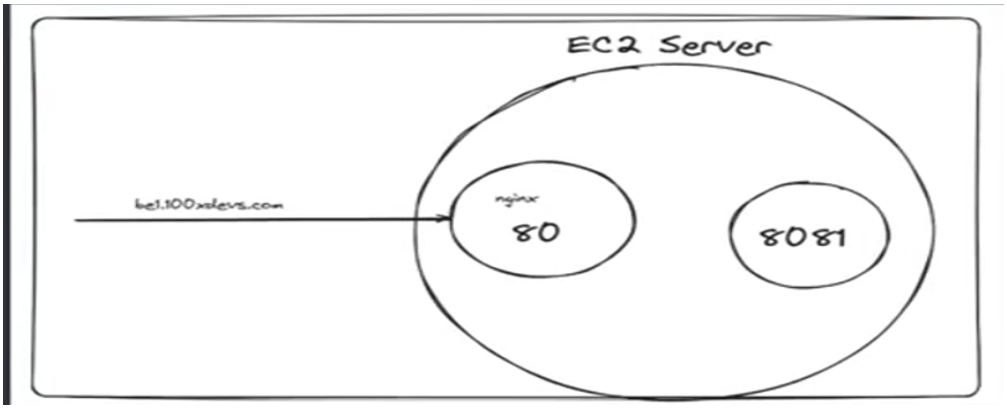
Installing Nginx

Run this command in the **ubuntu machine or inside your VM instance**

```
sudo apt update
sudo apt install nginx
```

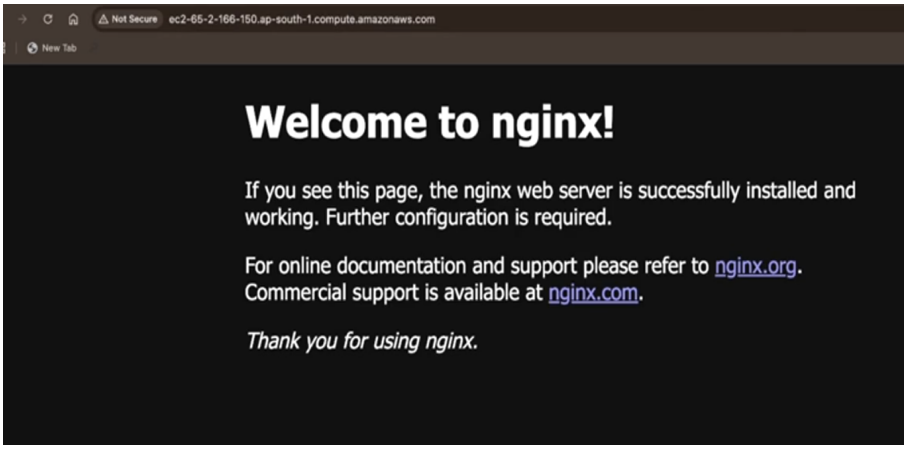
This should start a `nginx server` on port 80

Try visiting the website :-



> [!IMPORTANT] Remember to **open the port 80 if you have not by going inside the network security settings and adding the port 80 in the `Edit inbound rules > cutom TCP, anywhere` options**
>
> as you may have started the nginx but as it runs on port 80 so without opening it will not be able to do its work

and now if you go to the port 80 followed by the url of your website then you will see something like the below :-



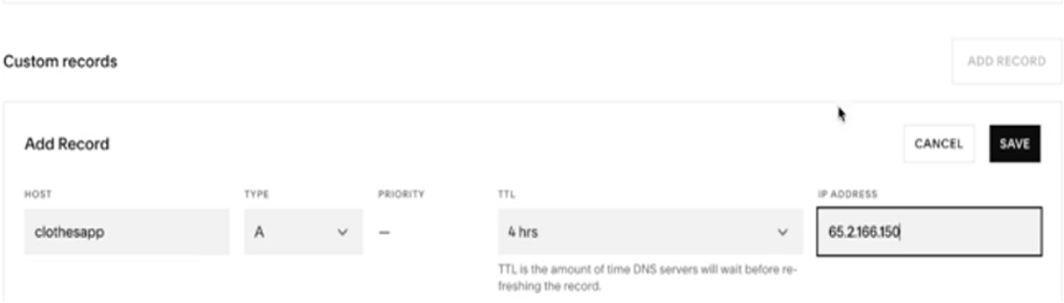
Further configuration is still required as you although have pointed port 80 to `nginx` but what about the next step which was to **point the nginx to a port on which your website is present**

So now **Go and explore the domain name**

-> Popular website for getting the domain name -> **Squarespace** and **Godaddy**

Setting the custom domain name from SquareSpace

Go to the `Add records` option and then just fill the required details of your website



Now sometimes it take time(even hours) to propagate this DNS.

How to check ?

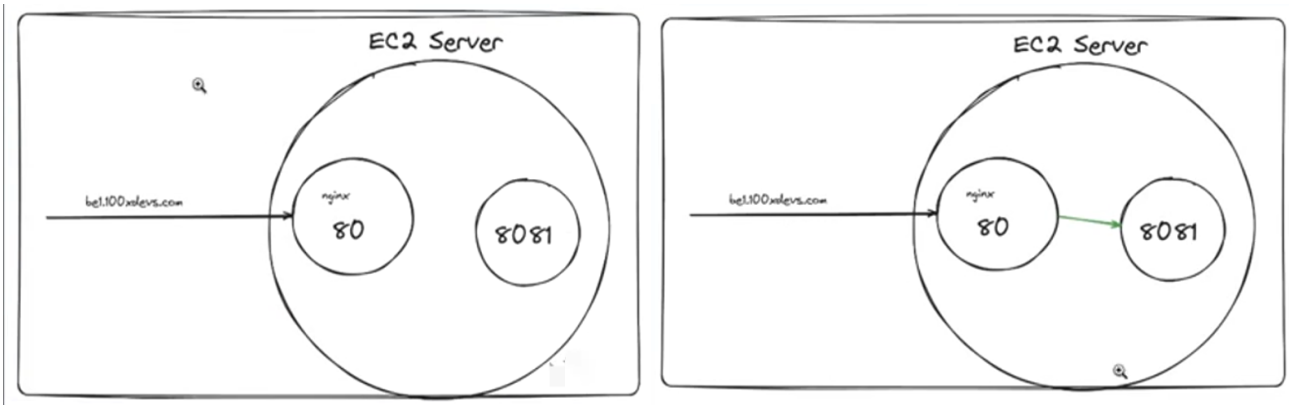
-> just run the below command

```
ping your_DNS_name
```

and if you see the ip then you are good to go, your website has propagated with this DNS.

Now you have to configure NGINX to point at these DNS you made

basically you have to go from (**left image below**) to (**right image below**)



you have to change the configuration of NGINX.

Now lets run now 2 node.js process on different ports (basically 2 different projects and with different ports on which they are listening at)

Now coding the logic part where you are going to configure the NGINX in such a way that **respective website respective port pr jaye**

> [!IMPORTANT] **All the configuration in the NGINX happens through a file present at**
>
> just run the below command to reach inside that file
>
> `sudo vi /etc/nginx/nginx.conf` // sudo means **Super User DO** as to achieve something like this you have to be Superuser

You will get some pre-dafault file so get rid of all of them and then inside it paste the below code

```
sudo rm sudo vi /etc/nginx/nginx.conf
sudo vi /etc/nginx/nginx.conf


events {
    # Event directives...
}
http { // Any http request
    server { // coming on the server
    listen 80; // on port 80
    server_name clothesapp.l0Oxdevs.com;  // whose domin name is
clothesapp.100xdevs.com

    location/{ // that should reverse proxy on ("/" basically means any request
coming to this port)
        proxy_pass http://localhost:8080; // port 8080
        // Everything below is used when you are working with the WEB SOCKET
(ignore for now)
        proxy_http_version 1.1;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection 'upgrade';
        proxy_set_header Host $host;
        proxy_cache_bypass $http_upgrade;
    }
    // basically the above code is saying that whenever any http request is coming
from port 80 with server name clothesapp.100xdevs.com forward it to the port 8080
and similar one is for the below another server made
    }
    // similarly you can add another server if you are using multiple projects
(just keep on adding sections as you are running multiple projects on the same VM
instance)
    server{
        listen 80;
        server_name shoesapp.100xdevs.com;
        location/{
            proxy_pass http://localhost:8081; // port 8081
            proxy_http_version 1.1;
            proxy_set_header Upgrade $http_upgrade;
            proxy_set_header Connection 'upgrade';
            proxy_set_header Host $host;
            proxy_cache_bypass $http_upgrade;
        }
    }
}

sudo nginx -s reload  // Reloads the Nginx
```

Now doing the above thing you were able to **create two clean urls without having the tension of giving their respective ports while using their URLs to visit the projects**

- **We just need to open the port 80 and automatically user will be able to redirect to the respective website as per the DNS it has given**

-> Now going by the above approach we have still one problem which is -> `http` instead of `https` and hence for including that in your project you need to learn **Certificate Management**

Link to documentation -> Certificate Management

But above it the **DUMB way to deploy using HTTPS**

Assignment

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1. Get a GCP account and do this there
2. Replace `nginx` with `traefic`, `HA Proxy`, `Apache`
3. Try deploying a react app
4. Get a domain (go to **Namecheap** [it has cheaper domain names])
5. Try **ASGs**(Auto scaling groups in AWS)
6. Try to do **certificate management yourself**
7. Forever or pm2 (process management) [now if you will shut down your laptop, everything will be gone away to deal with this process management is used]